

Customer Lifetime Value Prediction Model

Project Report | Submitted by: Trisha N

1. Introduction

Customer Lifetime Value (LTV) estimates the value a customer may generate over the relationship with a business. This project uses customer purchase-behavior features to predict LTV and create relative customer segments.

2. Abstract

The prepared dataset contains 1,200 customers. Missing numeric values were handled using median imputation and duplicate customer IDs were removed. A Random Forest regression model was trained using Frequency, Recency, Average Order Value (AOV), and Tenure. On the held-out test set, the model achieved MAE 113.56, RMSE 155.17, and R^2 0.911.

3. Tools Used

Python, Pandas, NumPy, Scikit-learn, Matplotlib, and Joblib. The project follows the internship brief, which specifies Python/Sklearn/XGBoost, Excel, feature engineering with Frequency, Recency and AOV, regression validation with MAE/RMSE, and LTV segmentation.

4. Steps Involved

- Import customer purchase-behavior data.
- Clean missing values and remove duplicate customer records.
- Prepare Frequency, Recency, AOV and Tenure features.
- Split data into training and test sets.
- Train a Random Forest regression model.
- Evaluate the model using MAE, RMSE and R^2 .
- Predict LTV for all customers and divide them into four relative LTV segments.
- Export the final prediction CSV and visualizations.

5. Key Results

Model MAE: 113.56. Model RMSE: 155.17. R^2 : 0.911. The strongest model features by Random Forest importance were: Average_Order_Value, Frequency, Tenure_Months, Recency_Days. The final CSV contains predicted LTV and an LTV segment for every customer.

6. Conclusion

The completed model demonstrates how customer purchase behavior can be transformed into predicted lifetime value. The resulting LTV segments can be used as an analytical input for targeted marketing and customer-retention planning. Predictions are model estimates and should be validated with real business outcomes before operational use.